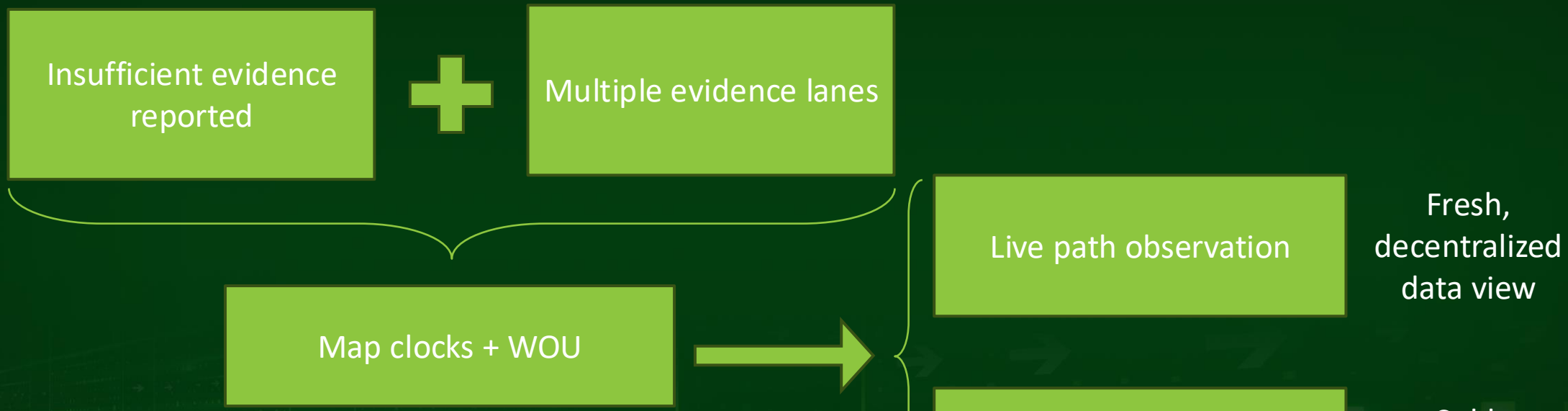
Clock-Aligned Telemetry for AI Clusters From BMC to GPU to Fleet



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turbalance

Overview


Insufficient evidence reported
Multiple evidence lanes
Map clocks + WOU
Live path observation
Forensic path observation
Fresh, decentralized data view
Cold, centralized data view

Dashboards still leave evidence gaps

- **Fragmented views: GPU counters alone do not explain host pressure, placement, fabric behavior, or power and thermal state.**
- **Mixed clocks: A shared dashboard window can still mix samples from different local timebases.**
- **Ambiguous absence:** A zero cannot safely stand in for an unsupported, inaccessible, or uncollected metric.
- **Missing context:** Benchmarks lose meaning without workload, topology, collector, and capability metadata.
- **Operator question: Where did useful compute go, and which signal is trustworthy enough to act on?**

Source: accepted session abstract supplied by the author.

Useful compute spans six evidence lanes

- **GPU:** DCGM, NVML, and gpustat utilization, health, memory, power, and link evidence.
- **Host:** CPU, memory, storage, kernel, process, and eBPF/OpenTelemetry-style pressure signals.
- **BMC:** Redfish power, thermal, sensor, inventory, and health state outside the host OS.
- **Placement:** Scheduler and Kubernetes job, pod, topology, and allocation context.
- **Fabric:** Network and NCCL behavior plus controlled benchmark context.
- **Time: PTP/NTP/chrony source, offset, skew, stratum, leap, lock state, time maps, and window of uncertainty.**

Sources: NVIDIA DCGM/NVML; DMTF Redfish Telemetry; chrony, linuxptp, RFC 5905, and Tangle.

Tangle turns clock state into event association

NODE A
local time 8.427 s
GPU stall observed
PTP locked • offset +8 µs
WOU ±12 µs



Raw local time misleads

NODE B
local time 8.570 s
Network pause observed
NTP slewing • offset +143 ms
WOU ±160 ms

Compare after mapping local time to UTC and carrying source, offset, freshness, and WOU

Illustrative timestamps only. Tangle maps local monotonic time to UTC; WOU expresses event-order confidence.

Evidence contract per sample

- **Identity:** source, host, device, interface, collector version, and capability profile.
- **Time: observed_at, received_at, clock source, offset, time-map ID, WOU, and sample freshness.**
- **Value:** metric name, unit, typed value, quality, and explicit support state.
- **Context:** job, pod, scheduler placement, topology, benchmark ID, and control state.
- **Provenance:** API or command, raw evidence reference, errors, transformations, and schema version.
- **Rule: unsupported or inaccessible stays explicit; never synthesize a plausible metric.**

Live path or forensic path?

LIVE PATH
1-second sampling target
GPU • host • BMC • scheduler • time map
Low-cost, continuous status
Freshness gates operator views

FORENSIC PATH
Triggered or scheduled
eBPF • NCCL • deeper probes
Tangle-style reconstruction
Higher cost, richer evidence

Both paths emit the same source-bundle and time-map contract

Collection strategy from the accepted abstract. Status, benchmarks, and time maps remain distinct data products.

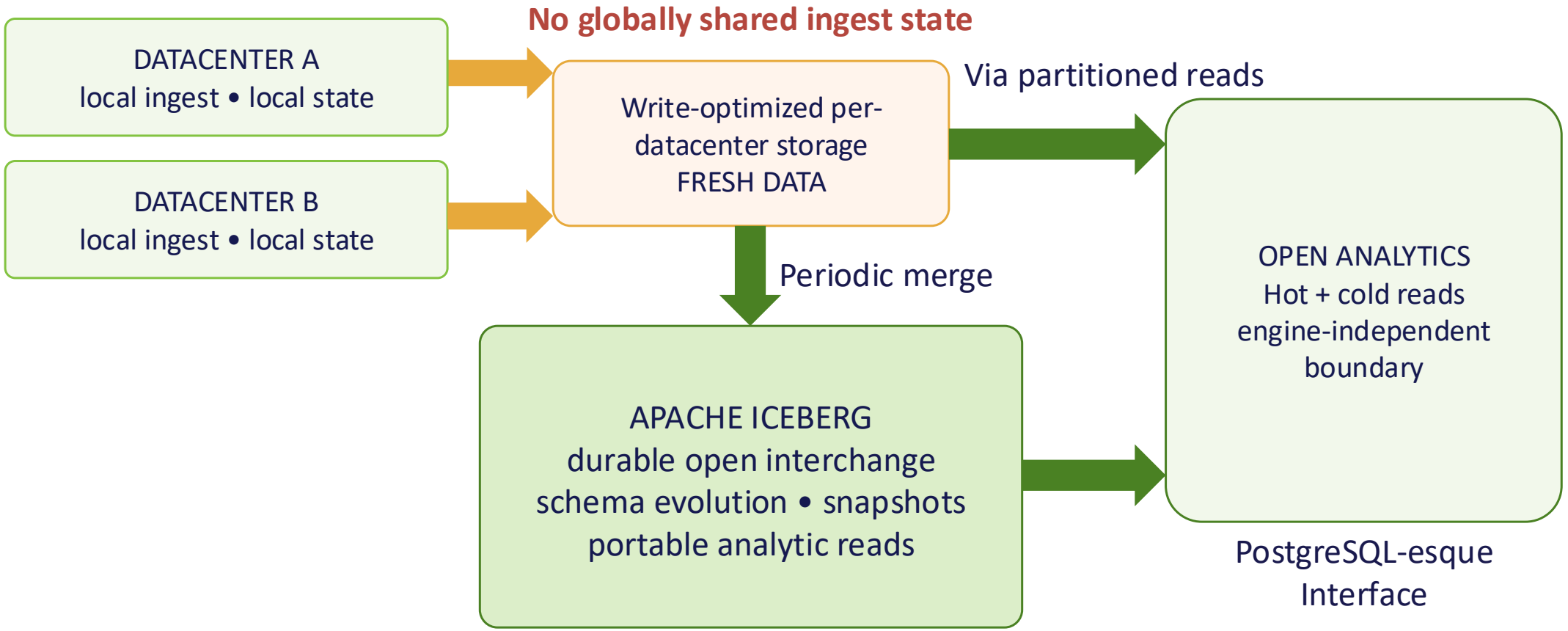
Agents report gaps, not guesses

- **Discover: inventory GPUs, BMC endpoints, interfaces, clock services, time maps, and scheduler context.**
- **Gate credentials: deploy privileged collectors only where endpoints and authorization exist.**
- **Advertise capability: supported, unsupported, inaccessible, disabled, or unavailable.**
- **Collect conservatively: preserve required precision, WOU, and provenance.**
- **Emit evidence: carry errors and source freshness beside values.**
- **Compare safely: cohort separately when a required capability is absent.**

Heterogeneity validates the contract

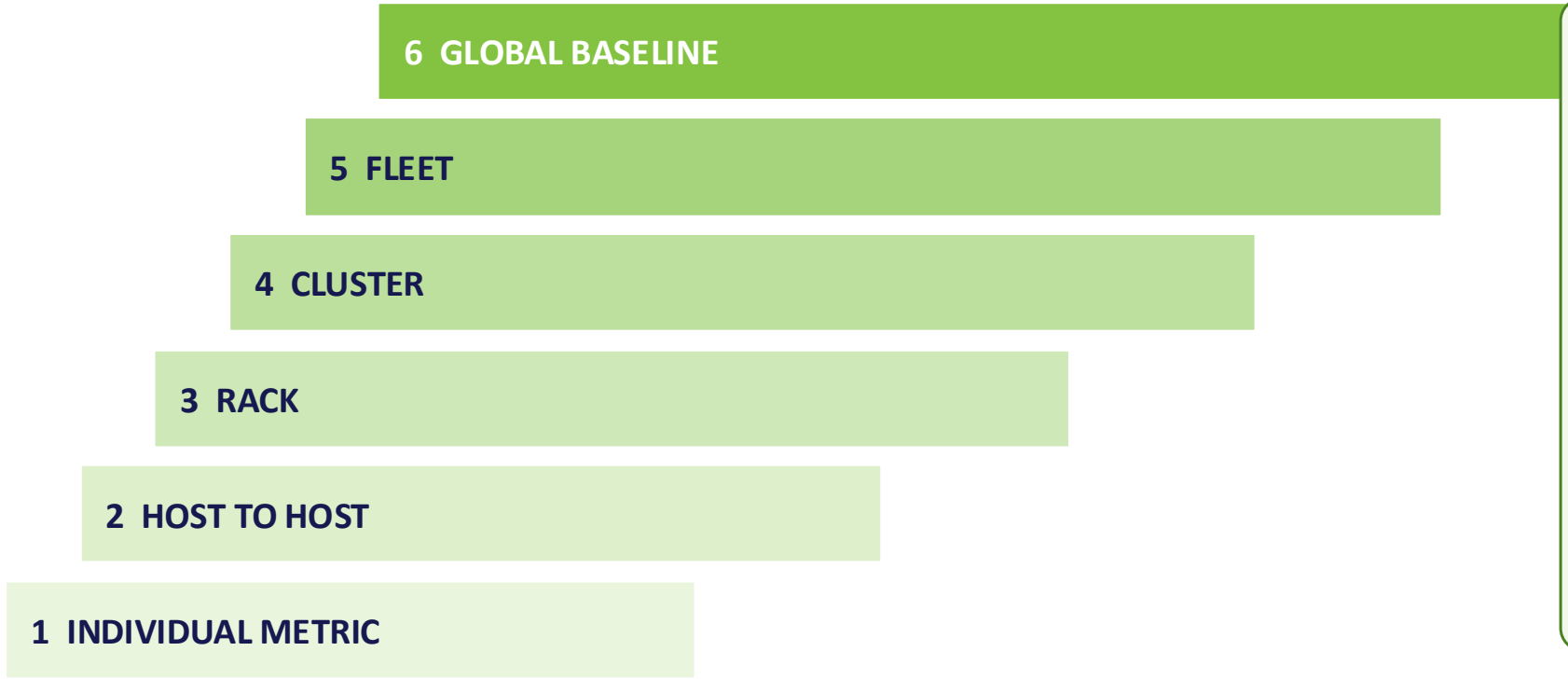
- **Accelerated hosts: DGX/GB10-class nodes exercise GPU, NVML/DCGM, NCCL, power, and thermal lanes.**
- **Controller appliance: coordinates discovery, enrollment, source bundles, and comparison jobs.**
- **Edge nodes: expose sparse telemetry, partial capability, and different clock paths.**
- **Infrastructure: Prometheus/Grafana, Kubernetes, Redfish, time maps, and PTP/NTP/chrony.**
- **Validation: prove behavior across unequal nodes, not production-scale benchmark results.**
- **Open posture: lab-built, non-confidential, and community-ready.**

Massive ingestion keeps local state local



Architecture from the accepted abstract. Iceberg references: iceberg.apache.org and the Apache Iceberg table spec.

Compare wider only when context matches



MATCH BEFORE EXPANDING
metric + unit
capability profile
clock window + uncertainty
workload / benchmark context
collector + schema version

Six-level comparison ladder from the accepted abstract. Broader baselines require stricter cohort controls.

Useful compute becomes a testable chain

Why is useful compute lower than expected?

LOCATE
Find where loss appears
GPU • host • BMC • fabric

TIME-ALIGN
Map local clocks to UTC
check freshness + WOU

CORROBORATE
Require a second lane
reject unsupported fields

COMPARE
Choose the narrowest valid benchmark cohort

ACT
Change only what the evidence can support

If provenance, freshness, support, or WOU fails: stop, disclose, and collect more evidence

Decision logic synthesized from the accepted abstract and the Tangle event-association model; no measured result is asserted on this slide.

Four gates decide if a signal is actionable

1

SUPPORTED
Does this node expose the metric?

2

FRESH
Is the sample inside the decision window?

3

TIME-VALID
Is clock state mapped and WOU acceptable?

4

TRACEABLE
Can the value be traced to raw evidence?

PASS ALL FOUR → SAFE TO COMPARE OR ACT

Actionability profile derived from the source-bundle, freshness, capability, clock-map, and WOU requirements.

Portable clock-aligned contracts

- 1. Source bundle: identity, time, value, context, provenance, and errors.
- 2. Time and capability: support state, freshness, clock map, WOU, uncertainty, and cost.
- 3. Comparison profile: cohort rules from individual metric through global baseline.
- Map, do not replace: Redfish, DCGM/NVML, Kubernetes, OpenTelemetry-style evidence, Tangle-style maps, and clock tools remain source interfaces.
- Interchange boundary: Iceberg carries durable open tables while hot stores remain implementation choices.
- Community outcome: fixtures expose unsupported evidence instead of hiding it.

OCP call to action

- **Convene: DCT, Hardware Management, Server/HPC, TAP, and AI HW/SW CoDesign contributors.**
- **First 30 days: agree on the minimal field dictionary and support-state vocabulary.**
- **By 90 days: publish sample bundles, clock/freshness/WOU rules, mapping fixtures, and conformance examples.**
- **Next milestone: map Redfish, DCGM/NVML, host, scheduler, network, Tangle-style maps, and time sources.**
- **Prove interoperability: run a multi-vendor lab demo with the six-level comparison ladder.**
- **Join: opencompute.org/community and the OCP Time Appliances Project community.**

Discussion

